



**Politecnico
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Politecnico di Torino

Lab report for the course Qubit Electronics

Master degree in Quantum Engineering

Group 1

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CHAPTER 1

Modified Code

Question: Report the modified code commenting your modifications.

```
1 num_states = 5
2
3 # Instantiate Schrodinger solver's parameters
4 params_schrod = SchrodingerSolverParams()
5 params_schrod.num_states = num_states      # Number of states to consider
6 params_schrod.tol = 1e-9
7
8 # Get the potential energy from the band edge for usage in the Schrodinger
9 # solver
10 d.set_V_from_phi()
11
12 # Create a submesh including only the dot region
13 submesh = SubMesh(d.mesh, dot_region)
14
15 # Create a subdevice for the dot region
16 subdvc = SubDevice(d, submesh)
17
18 # Create a Schrodinger solver
19 schrod_solver = SchrodingerSolver(subdvc, solver_params=params_schrod)
20
21 # Solve Schrodinger's equation
22 schrod_solver.solve()
23
24 # Output and save single-particle energy levels
25 subdvc.print_energies()
26 energies = subdvc.energies
27 np.save(path_out/"qdfdsol_energies.npy", energies)
```

First of all, we set the number of energy levels to consider equal to 5. Next, we instantiated the Schrodinger solver's parameters and set the tolerance for the convergence to 1e-9. The potential to use for reference was defined. Then, we specified the region of the dot for which to solve the Schrödinger equation. We also added the commands for printing and saving the single-particle energy levels.

```
1 # Schrodinger
2 out_dict = {"Ground": np.abs(subdvc.eigenfunctions[:,0])**2,
3             "1st excited": np.abs(subdvc.eigenfunctions[:,1])**2,
4             "2nd excited": np.abs(subdvc.eigenfunctions[:,2])**2,
5             "3rd excited": np.abs(subdvc.eigenfunctions[:,3])**2,
6             "4th excited": np.abs(subdvc.eigenfunctions[:,4])**2,
7             "conf-potential": subdvc.V/ct.e}
8 io.save(str(path_vtu_q), out_dict, subdvc.mesh)
```

```
9 with open(path_energies, 'w') as f:  
10     f.write('Energy eigenvalues [eV]')  
11     f.write(str(subdvc.energies/ct.e))  
12  
13 subdvc.print_energies()
```

This piece of code computes the probability densities by squaring the absolute values of the eigenfunctions corresponding to the first five energy levels (ground state and four excited states). It also extracts the confining potential profile and converts the energy eigenvalues from joules to electronvolts (eV). The results, including the probability densities, potential, and energy eigenvalues, are saved.

CHAPTER 2

Results

2.1 Energy Eigenvalues

The energy eigenvalues, obtained from the simulation, are the following:

- **ground state:** 0.01128982 eV;
- **first excited:** 0.01129278 eV;
- **second excited:** 0.01407358 eV;
- **third excited:** 0.01408196 eV;
- **fourth excited:** 0.0144833 eV.

One can observe that some levels are close to each other:

- The ground and 1st excited are separated by a splitting of ($\sim 3 \mu\text{eV}$);
- The 2nd and 3rd excited have a slightly larger splitting ($\sim 8 \mu\text{eV}$).

These closely spaced levels are consistent with bonding (symmetric) and antibonding (antisymmetric) states formed due to tunnel coupling between the two dots. If the dots were completely uncoupled, one would see perfect degeneracy: each level would appear twice, once in each dot. When the barrier between the dots is finite, an electron can tunnel between the dots. This tunneling coupling lifts the degeneracy, leading to the formation of the observed symmetric and antisymmetric states.

2.2 Conduction Band

The file `DQP_SiMOS_EC.vtu` contains the data about the resulting conduction band. In the following figure, we report the plot obtained.

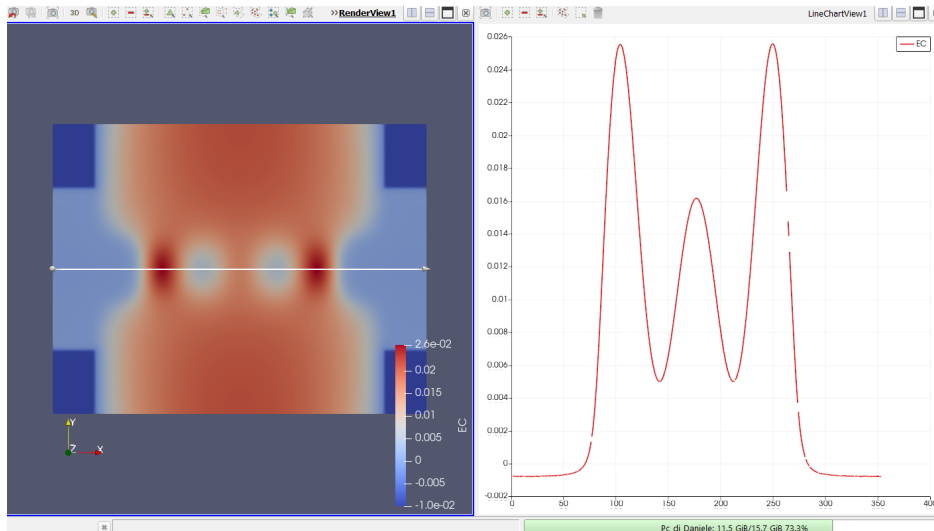


Figure 2.1

The behavior of the conduction band is unvaried with respect to the previous lab. And this is expected since we did not change the parameters of the Poisson solver. The only difference is the added line in the simulation

```
1 # Get the potential energy from the band edge for usage in the Schrodinger
2 # solver
3 d.set_V_from_phi()
```

that specifies that the potential to use in the Schrödinger equation is the one resulting from the solution of the Poisson equation. From the documentation, we read that when solving the Poisson equation, `d.phi` (the electrostatic potential) is correctly updated. However, QTCAD does not automatically update the internal fields used for the quantum potential, such as `V`, `EC`, and `EV`, unless the method `d.set_V_from_phi()` is explicitly called. If this step is omitted, functions like `d.cond_band_edge()`, which are used to generate the `EC.vtu` file—may operate on an outdated or inconsistent potential field. As a result, the extracted conduction band edge can appear noisy, discontinuous, or physically incorrect, which is clearly observable in the corresponding plot.

2.3 Wavefunctions

We report the plot of the wavefunctions of the ground and first excited level. As we expected, they exhibit respectively a symmetric and an antisymmetric function.

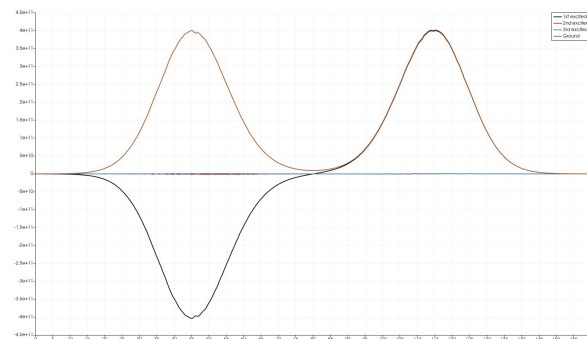
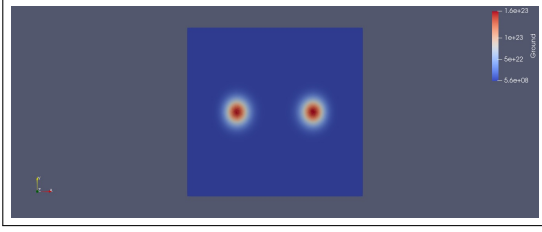
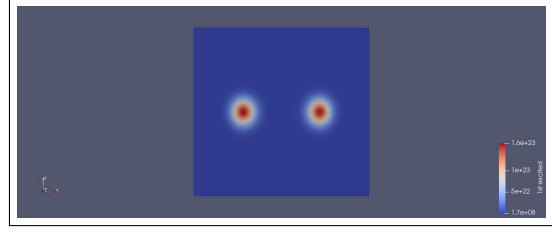


Figure 2.2

This behavior is further supported by the probability density functions, reported in the following heatmaps.



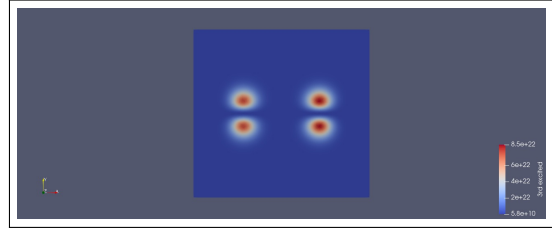
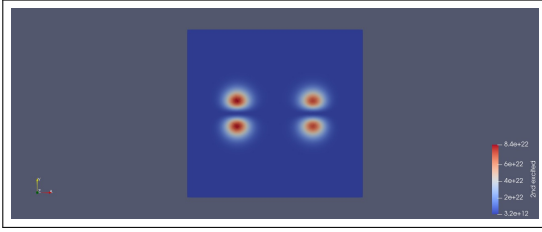
(a)



(b)

The ground state probability density exhibits two well-localized maxima corresponding to the left and right quantum dots. The absence of a node between them confirms that this state is symmetric (bonding), resulting from the quantum mechanical coupling between the two dots. This delocalized wavefunction indicates coherent tunnel coupling and energy splitting of the two levels.

The first excited state shows identical spatial localization of the ground state, but differs by its wavefunction symmetry. As confirmed by the raw wavefunction plot, this state is antisymmetric, with a node at the interdot barrier. It corresponds to the antibonding molecular state, formed due to quantum tunneling between the dots.



The same can be observed for the other two paired levels (2nd and 3rd excited). They show a clear four-lobe structure, indicating excited orbitals within each quantum dot, likely along orthogonal directions.

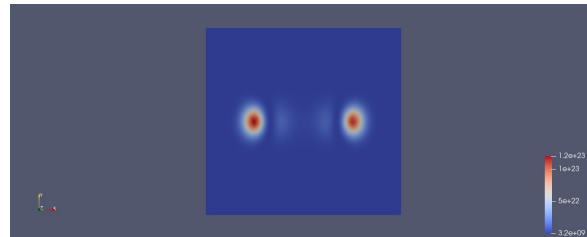


Figure 2.5

The fourth excited state again displays a symmetric localization on the two dots but with higher energy and a more delocalized shape, due to a more complex wavefunction.